

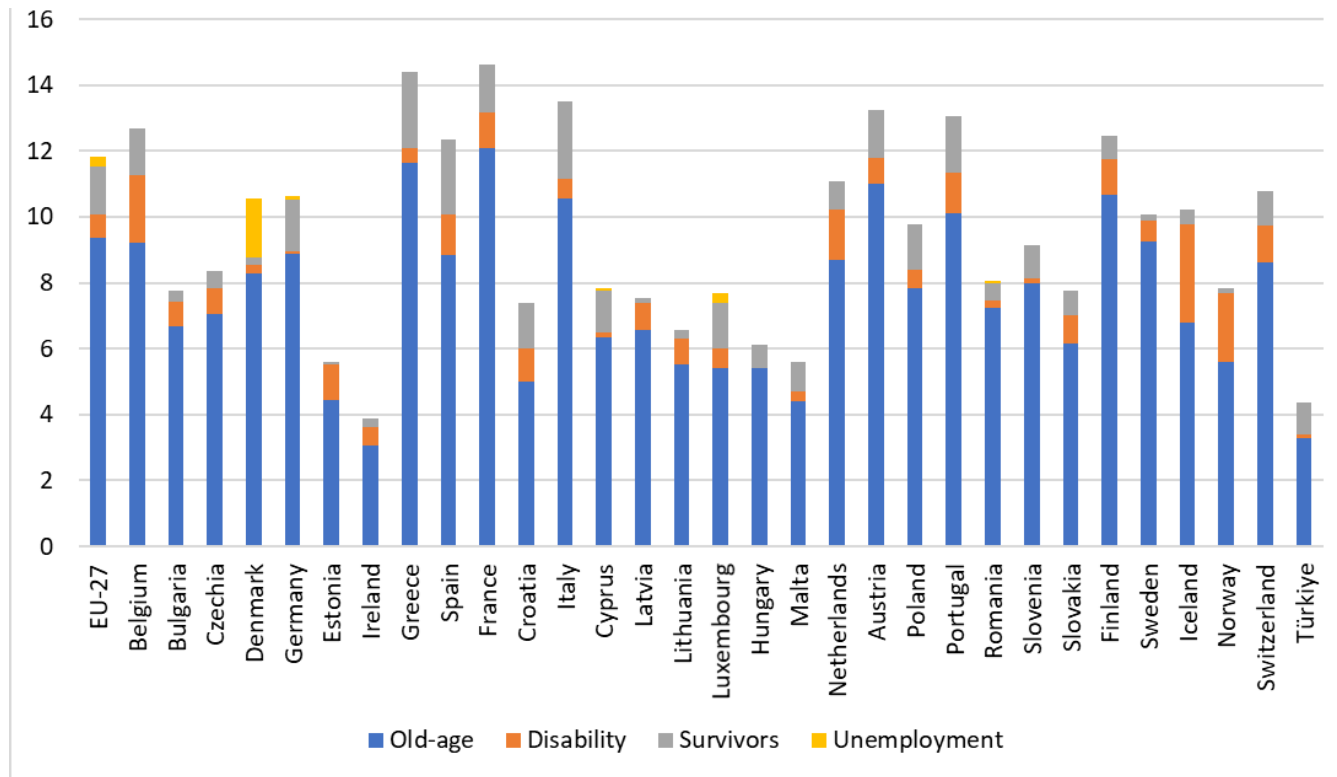
30264 - Public Finance

BIEM
Spring 2026

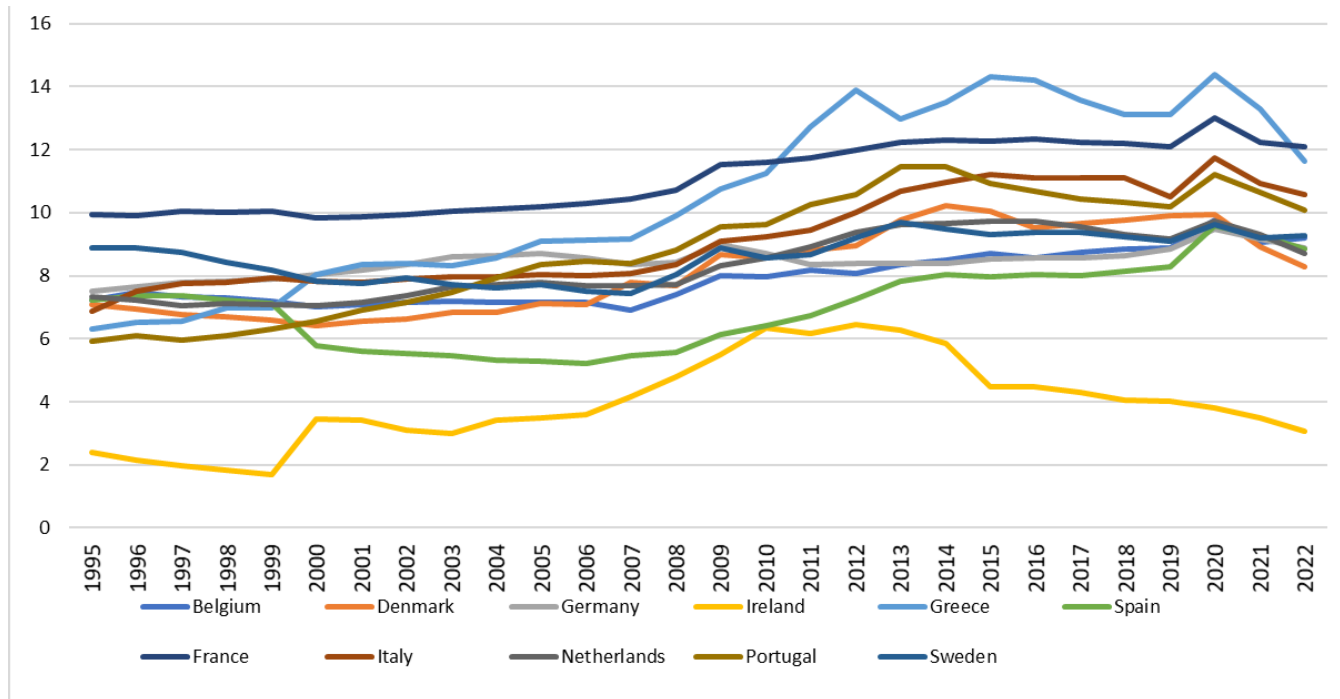


Social Security (Pension System) **Gruber - Chapter 13**

Social Security spending in Europe (% of GDP)



Pension spending trends in Europe



In this slide set

- Why should the government provide social security?
- Types of pension systems («Pay-as-you-go» vs «fully funded»)
 - Funding scheme
 - Comparison
- Public pension system' risks (moral hazard, crowd-out) and benefits
- Financial equilibrium and social security reforms
- A focus on the Italian case

Why should the govt. provide social security?

- **Adverse event:** being too old to gain a sufficient income
 - **Market solution:** Private pension plans: savings today are invested in assets such as government bonds, corporate bonds, or stocks, and the accumulated assets pay for (fund) the future benefits promised by the pension (**fully funded** plans).
 - **Problems of private pension plans:**
 - ✓ poor performance of financial markets and inflation can lead to low future pensions → the ratio of benefits received to earnings prior to retirement (so called **replacement ratio**) may be too low;
 - ✓ people under-estimate their future needs, or think they can work forever...
- Private pension plans can be an inefficient way to deal with the social risk of losing the ability to gain an income.

Types of pension systems

- **Fully funded:** social security contributions paid by workers are invested in the capital market in *assets* that will fund their future pensions.
- **PAYGO (Pay-as-you-go):** the revenue collected in each period is devoted to the funding of the pensions paid in the same period.



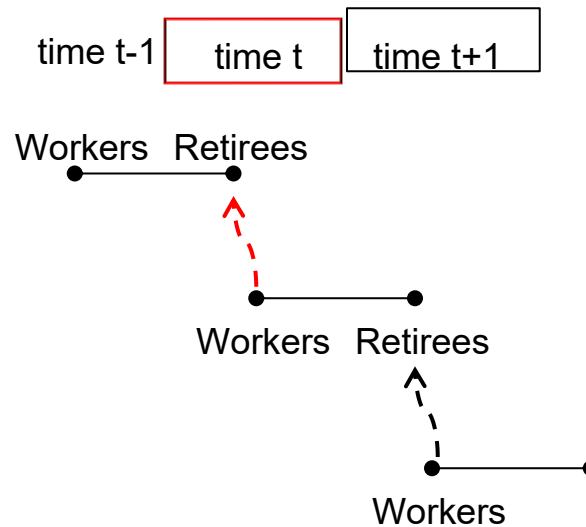
Public pension systems

- Public Social Security has traditionally been an **unfunded** plan: taxes or social contributions collected from a worker today go directly to today's retirees rather than being saved (and invested) to pay for benefits when the taxed worker retires.
- This is often referred to as a **pay-as-you-go** scheme and it is based on an inter-generational pact: active workers pay for the pensions of today's retirees because they know that when they will stop working the next generation will pay for their pensions.

How are Pension Systems funded?

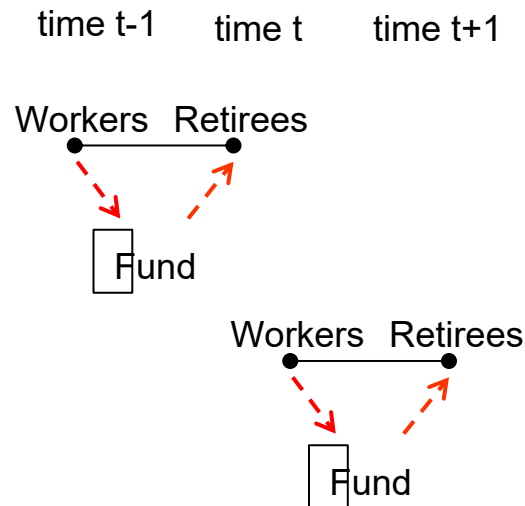
➤ Pension Systems differ in their funding:

➤ **Pay-As-You-Go (Unfunded)**



How are Pension Systems funded?

- Pension Systems differ in their funding:
 - Pay-As-You-Go
 - **Fully Funded**



Comparison

We want to compare these two systems over three main dimensions:

- 1) Financing method
- 2) Rate of return
- 3) Method for calculating the pension benefit

Comparison

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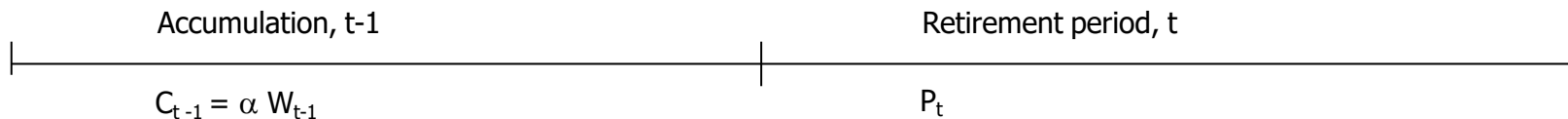
1) Financing method

2) Rate of return

3) Method for calculating the pension benefit

Financing Method

Let's consider 2 periods (working & retirement periods).



➤ **Pay-go system:** the revenue collected in each period (the sum of the social security contributions paid) is used to fund the pensions paid in the same period:

- $C_{t-1} \Rightarrow P_{t-1}$ and $C_t \Rightarrow P_t$

➔ «**INTERGENERATIONAL AGREEMENT**»

➤ **Fully funded system:** social security contributions paid by each worker during her/his working period are invested in the capital market. The pension he/she will receive will be equal to the contributions paid augmented by the return i (*interest rate*) obtained in the capital market.

- $P_t = C_{t-1}(1+i)$

- Accumulation of resources: **insurance perspective**

Comparison

We want to compare these two systems over three main dimensions:

1) Financing method

2) Rate of return

3) Method for calculating the pension benefit

Pension systems' returns

Let's consider 2 generations:

- The former works at time $t-1$ and is on retirement at time t ;
- The latter works at time t and will on retirement at time $t+1$.

Let's define:

- Number of workers: N_t and N_{t-1}
- Average wage of the workers: w_t and w_{t-1}
- Total wage $W_t = N_t w_t$ and $W_{t-1} = N_{t-1} w_{t-1}$
- Contribution rate (proportional to wages): α

Suppose that:

- $w_t = w_{t-1}(1+m)$ with m = productivity growth rate
- $N_t = N_{t-1}(1+n)$ with n = population growth rate (assumed to be equal to the workers growth rate)

Hence:

$$W_t = N_t w_t = N_{t-1}(1+n)w_{t-1}(1+m) = W_{t-1}(1+n)(1+m)$$

Pension systems' returns

- **Paygo system:** the total amount of pensions paid at time t is equal to the total amount of contributions paid at time t

$$P_t^{PG} = \alpha W_t = \alpha W_{t-1}(1+n)(1+m)$$

- **Fully funded system:** the total amount of pensions paid at time t is equal to the total amount of contributions paid at time $t-1$ plus the return obtained on the capital market, which is equal to the market interest rate i

$$P_t^{FF} = \alpha W_{t-1}(1+i)$$

- In the previous formula, the rates $(1+n)(1+m)$ and $(1+i)$ can be interpreted as the return rates of the social contributes paid during the working life (period $t-1$) by the individuals that will be retirees at time t .

Pension systems' returns

- More precisely, we define «**implicit return**» that the pension system guarantees to a given generation the ratio between the total amount of pensions paid by the State and the total amount of social contributions paid by workers, decreased by 1:

$$(P_t/C_{t-1})-1$$

- $(P_t^{PG} / \alpha W_{t-1})-1 = (1+n)(1+m)-1 = 1+n+m+nm-1 \approx n+m$ (paygo implicit return rate)
 - $(P_t^{FF} / \alpha W_{t-1})-1 = (1+i)-1 = i$ (fully funded return rate)
- Paygo implicit return rate is higher than fully funded return rate if:

$$(n+m) > i$$

That is if the sum of population growth rate and wages growth rate (= productivity growth rate) is greater than market interest rate i .

Paygo system return rate: example

- Population and earnings grow by 5% per year.
- Unfunded system with 10% payroll tax.

Period	Number of Young Workers	Earnings per Young Worker	Taxes Paid by Young Worker	Total Taxes Paid	Number of Old Retirees	Benefits to Old Retirees	Taxes Paid by Old Retirees	Rate of Return
1	100	\$20,000	0	0	0	0	—	—
2	105	\$21,000	\$2,100	\$220,500	100	\$2,205	0	Infinite
3	110	\$22,050	\$2,205	\$242,550	105	\$2,310	\$2,100	10%
4	115	\$23,153	\$2,315	\$266,225	110	\$2,420	\$2,205	10%
5	121	\$24,310	0	0	115	0	\$2,315	-100%

Paygo system return rate: example - Conclusions

- PRO: it is possible to start immediately the pension transfers to old individuals (retirees) even if contributions were not previously payed.
- FIRST GENERATION EFFECT: the PAYGO pension system induces, at the moment of its introduction, a **redistribution** in favour of the generation working before its introduction: the return rate tends to infinity for the first generation.
- LAST GENERATION EFFECT: if workers stop financing the system, retirees will get no pensions even if they contributed to the pension system during their working life: negative return rate for the last generation.



DIFFERENCES WRT THE FULLY FUNDED SYSTEM

Comparison

We want to compare these two systems over three main dimensions:

1) Financing method

2) Rate of return

3) Method for calculating the pension benefit

PAYGO pension system: alternative computation methods

Let's consider a PAYGO system. How can be pensions computed?

- 1) **Earnings-based:** the pension received each year is equal to a share (*coefficient of return*) of the pensionable earnings multiplied by the years of contribution.
- 2) **Contribution-based :** ratio of the sum of the contributions payed by the worker during her working life over the expected lenght of the retirement period.

Earnings-based PAYGO system

Define:

- β = coefficient of return;
- w_p = pensionable earning;
- L = years of contribution

Using the earnings-based method the pension is:

$$P = \beta \times w_p \times L$$

i.e. the pension is equal to the coefficient of return applied to the pensionable earnings, multiplied by the number of years in which the individual has paid contributions.

Earnings-based PAYGO system

The pensionable earning can be defined in different ways

1. w_p is equal to the **last salary** (w_L) before retirement:

$$P = \beta \times w_L \times L$$

2. w_p is an **average of the salaries** received during the working life augmented by a revaluation rate r set by the government.

Example with $L=3$:

$$w_p = \frac{w_1(1+r)^2 + w_2(1+r) + w_3}{3}$$

Contribution-based PAYGO

The pension is computed using the following equation:

$$C = AV(P)$$

where:

- C = sum of the contributions paid during the working period augmented by a revaluation rate r set by the government.
- $AV(P)$ = actual value of the pension flow computed at the end of the working period based on life expectancy

Example with $L=3$ (earnings: w_1, w_2, w_3)

$$C = \alpha w_1(1+r)^2 + \alpha w_2(1+r) + \alpha w_3$$

where α is the contribution rate paid each year by the worker

Contribution-based PAYGO

- Suppose that the pension P is constant over time (but there might be indexation...)
- Define $e(L)$ the years of life expectancy at the day of retirement.
- The actual value of the pension at the day of retirement if $e(L)=2$ is:

$$AV(P) = \frac{P}{(1+r_z)} + \frac{P}{(1+r_z)^2}$$

where r_z is the discount rate of future pensions.

Contribution-based PAYGO

If $r_z=0$:

$$AV(P) = e(L) \cdot P$$

$$C = AV(P) \rightarrow C = e(L) \cdot P \rightarrow P = \frac{C}{e(L)}$$

Pension P:

- Increases with C, and hence with α , r , w and L .
- decreases with $e(L)$

$1/e(L)$ = «transformation coefficient»: it is the coefficient that converts C into P. It reflects the real life expectancy, it allows tackling the ageing of the population transferring the burden on the retirees

Fully funded pension system: computation method

The pension is computed using the following equation:

$$C = AV(P)$$

where:

- C = sum of the contributions payed during the working period augmented by the interest rate i obtained on the capital market.
- $AV(P)$ = actual value of the pension flow computed at the end of the working period based on life expectancy

Example with $L=3$ (earnings: w_1, w_2, w_3)

$$C = \alpha w_1(1+i)^2 + \alpha w_2(1+i) + \alpha w_3$$

where α is the contribution rate payed each year by the worker

$$\Rightarrow AV(P) = \frac{P}{(1+rz)} + \frac{P}{(1+rz)^2} \quad \text{If } r_z=0 \Rightarrow P = \frac{C}{e(L)}$$

Summing up:

	PAYGO – Earnings- based	PAYGO - contribution-based	Fully-funded
Financing	Contributions paid by active workers (intergenerational agreement)	Contributions paid by active workers (intergenerational agreement)	Contributions payed by the retirees when they were workers
Pension Bill	Depends on: wages, contribution rate, productivity and number of employed	Depends on: wages, contribution rate, productivity and number of employed	Depends on: wages, contribution rate and capital market interest rate
(Individual) Pension	Depends on pensionable earnings, years of contribution and coefficient of return	Depends on: - Contributions payed during the entire working life - Contribution revaluation rate (r) - years of life expectancy at the day of retirement	Depends on: -Contributions payed during the entire working life - Capital market interest rate - years of life expectancy at the day of retirement

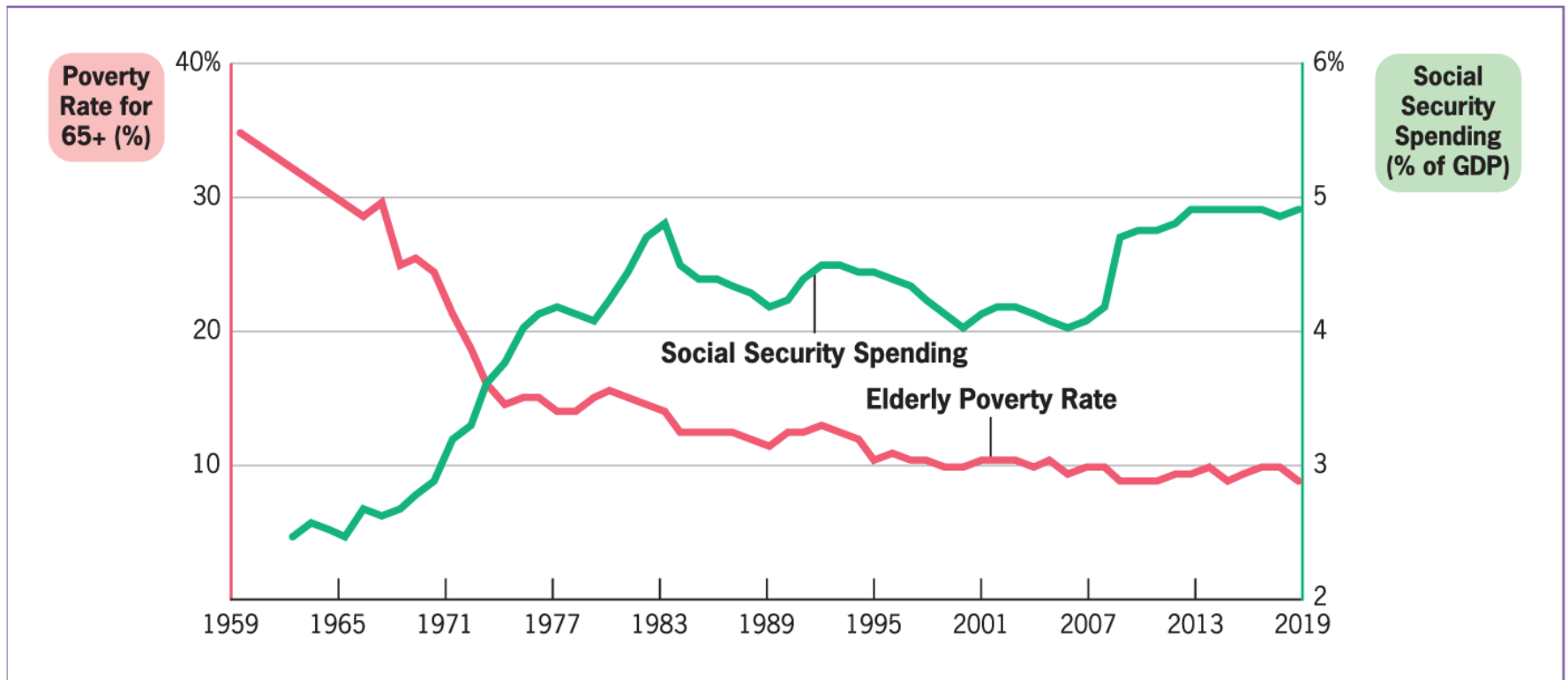
Justifications for State intervention in the pension system

- 1) Insurance market may experience market failures because of adverse selection (individuals have more information about their life expectancy than insurance companies)
- 2) Policy makers are afraid that people do not save enough during their working life for their retirement period (*paternalism*).

Issue: does the public pension system realizes consumption stabilization or it simply crowd-out private saving?

Living standards of the elderly

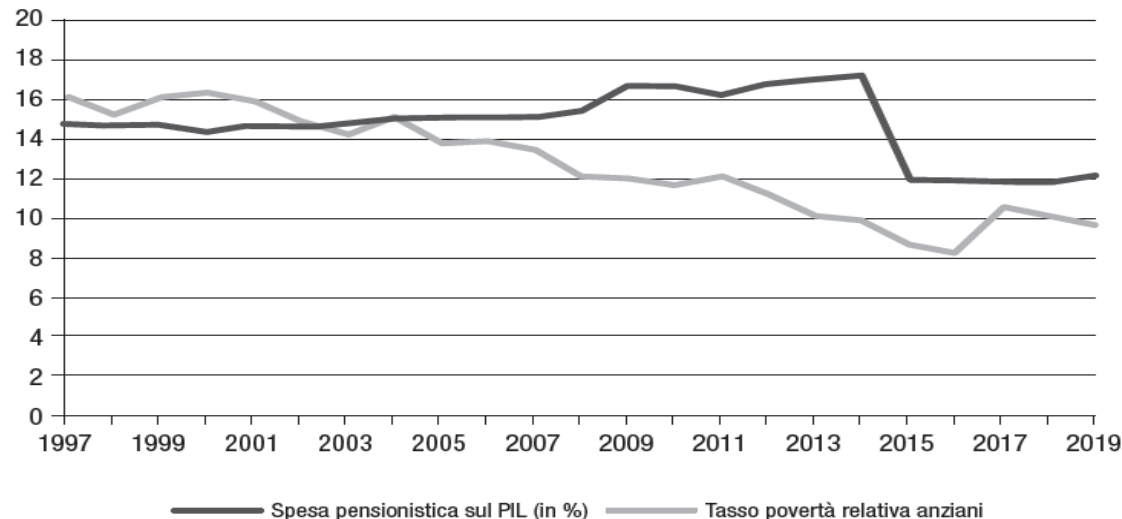
In the US, there is a striking negative correlation over time between the poverty rates of the elderly (which have fallen) and the size of the Social Security program (which has risen).



Living standards of the elderly

In Italy, we observe a correlation between the reduction in the relative poverty rate among the elderly and the increase in pension benefits; this would seem to confirm the hypothesis of a positive effect on the stabilization of consumption.

Figura 9.2 Povertà della popolazione anziana e sistema pensionistico in Italia (1997-2019)



Note: nel corso del tempo si osserva una correlazione inversa tra il tasso di povertà degli anziani (che diminuisce) e la dimensione della spesa pensionistica sul PIL (che aumenta).

Fonte: elaborazione su dati ISTAT (*Rilevazione sulle forze di lavoro*, 2022).

The impact of the pension system on the choice of retirement age

- The pension system guarantees an income to the retirees.
- It may cause a **moral hazard** problem: a worker may have an incentive to move up the beginning of the retirement period in order to benefit from the pension.

Why can the pension system influence the retirement decision?

«**Implicit taxation**» on labour → Incentive to move up the retirement date. The marginal cost of working one additional year (pay contributions for one additional year and one year less of pension benefit) is greater than the marginal benefit (receive a higher pension)

Social Security and Retirement: Theory

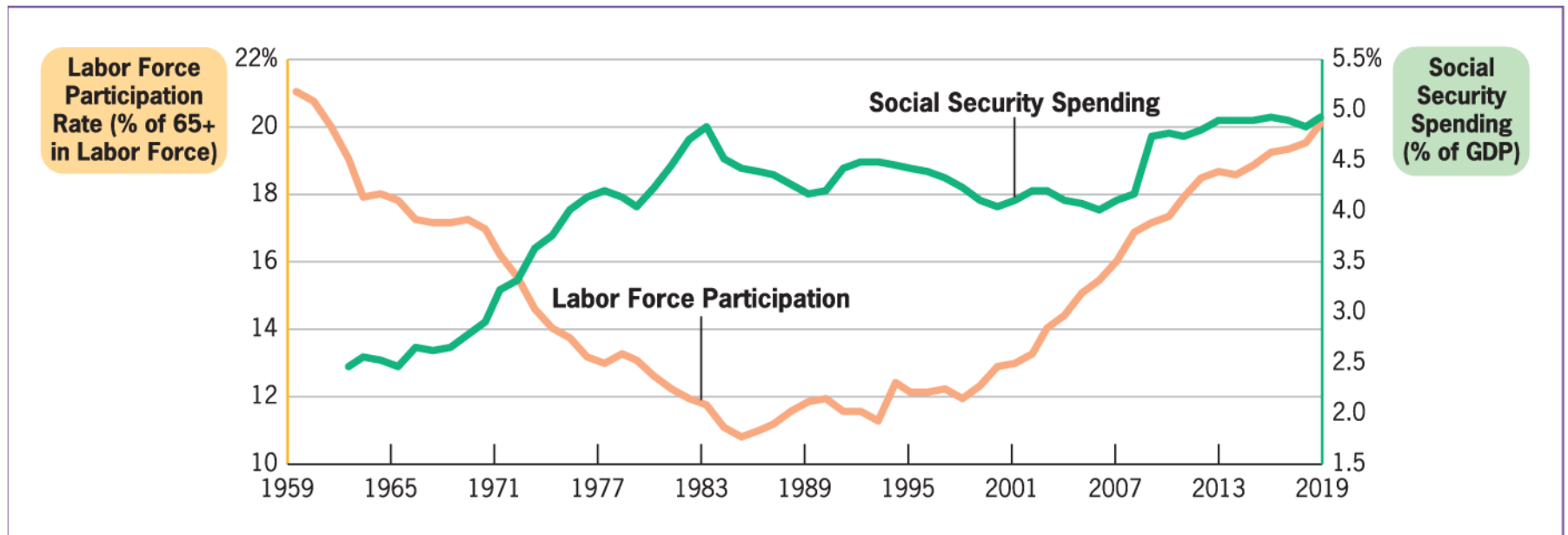
What are the costs and benefits of working an additional year at age 62?

- Costs: Implicit tax
 - Pay an extra year of payroll taxes on earnings.
 - Receive one year less of Social Security benefits.
- Benefits: Benefit adjustment
 - Higher Social Security benefit level through the actuarial adjustment
 - Increase pension

Moral Hazard: Evidence

Elderly Work and Social Security, US 1959–2019

Moral hazard: once pension program in place workers may retire earlier to start collecting their benefits

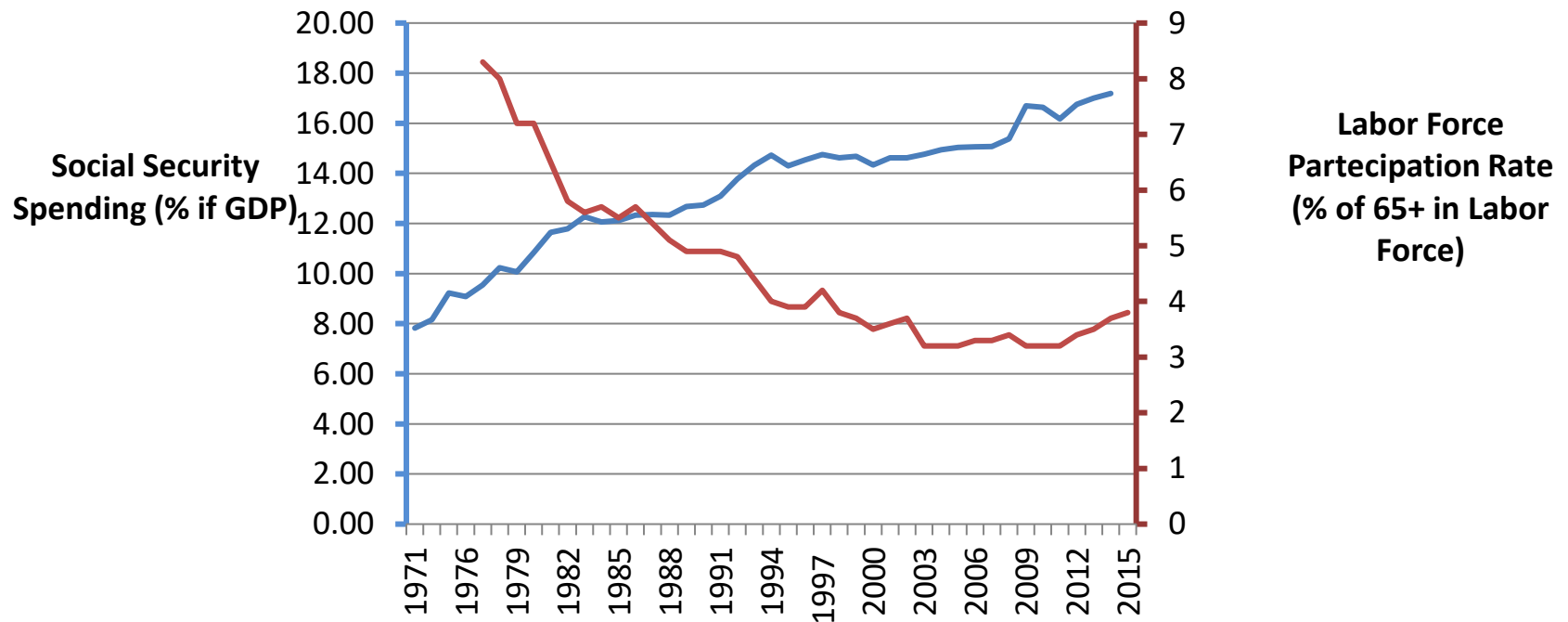


In the 1960s and 1970s, the program grew rapidly and LFP decreased. Both flattened in the mid-1980s, and in recent years, both are on the rise.

Moral Hazard: Evidence

Elderly Work and Social Security, Italy 1971–2015

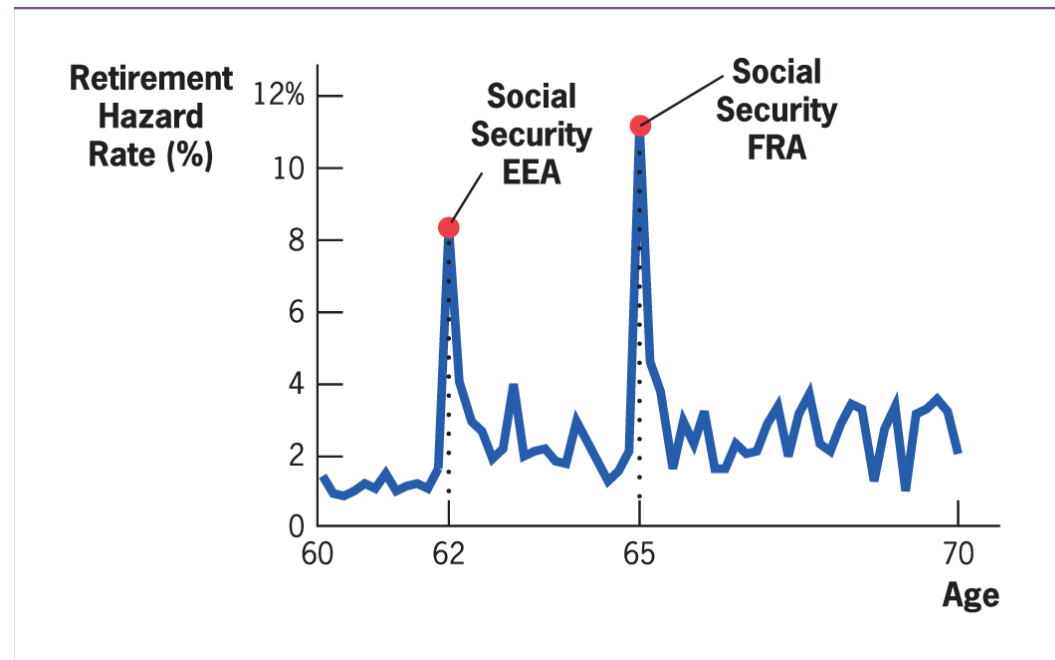
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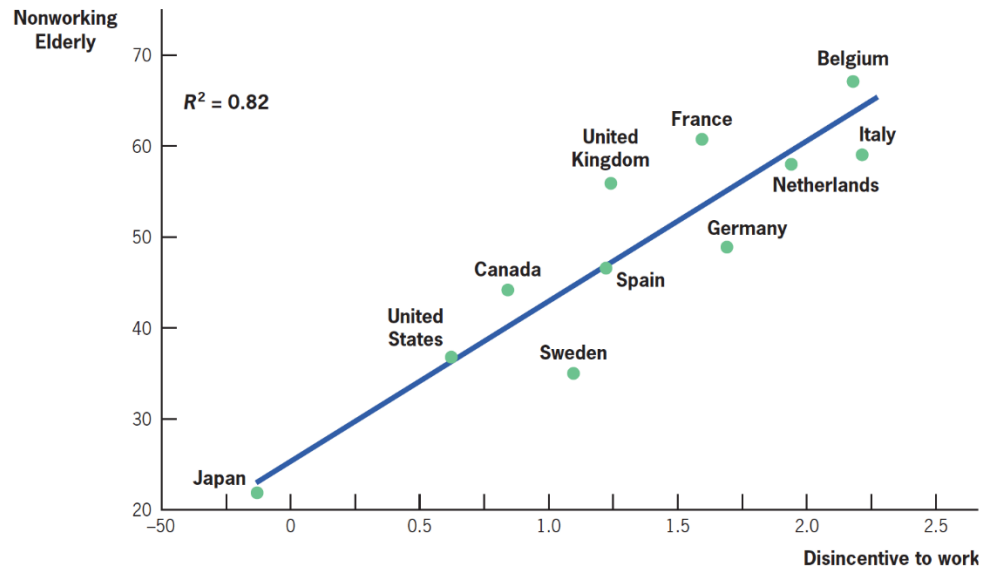
Social Security and Retirement: Evidence

Spike in Retirement Hazard at EEA

The male hazard rate, or exit rate at each age, given that a man has worked up to that age, spikes at ages 62 (the Early Entitlement Age, EEA) and 65 (the previous Full Retirement Age, FRA)—key ages for the Social Security system.



Implicit Social Security Taxes and Retirement Behavior



There is large variation across nations in the social security disincentives to work at older ages. Those nations with greater disincentives to work tend to have much higher nonwork among older workers.

Implications

- Evidence suggests that it is potentially very costly to design Social Security systems that penalize additional work beyond the retirement age.
- Adjusting systems to more fairly reward work at old ages can mitigate much of the moral hazard effect of Social Security.

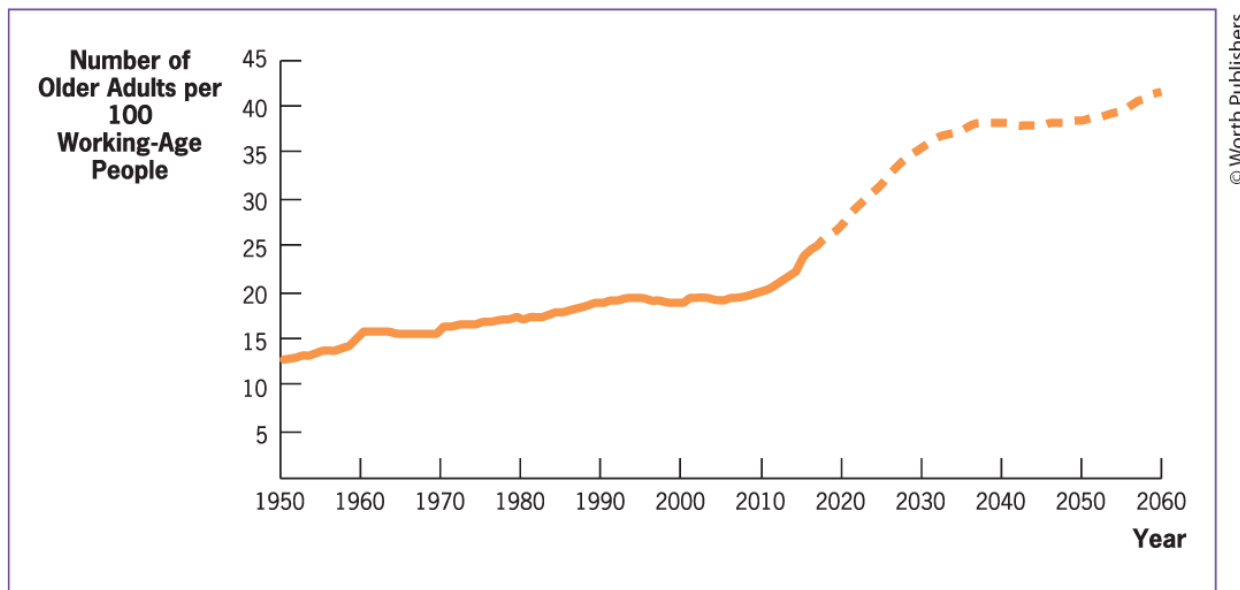
Social Security Reforms

Social Security faces a major fiscal imbalance as it is increasingly difficult for young generations to pay for the benefits of older generations.

- Rising life expectancy
- Falling birth rates
- Reduction in wage growth rates

The demographic dangers of pay-as-you-go pension systems

In the US, the number of persons over age 65 per working-age person age 15 to 64 more than triples over the century, from 13 per 100 in 1950 to 42 per 100 in 2060.



The financial equilibrium in a pay-as-you-go pension system

A pay-as-you-go system is in financial equilibrium when total pension expenditures are exactly equal to total contribution revenues. This condition can be written as:

$$P \cdot N_r = \alpha \cdot N_w \cdot w$$

where:

- P is the average pension,
- N_r is the number of retirees,
- N_w is the number of workers,
- w is the average wage,
- α is the payroll tax rate.

Therefore, financial equilibrium in a PAYG system depends on **demographic factors** (the ratio of workers to retirees), **average wages**, and **pension levels**.

The financial equilibrium in a pay-as-you-go pension system

$$P \cdot N_r = \alpha \cdot N_w \cdot w$$

If the population ages, N_r increases. To keep the system in balance, a reform of the pension system may:

- Increase α : workers pay higher contributions
- Reduce P : pensioners receive lower pensions
- Increase the retirement age: shifting individuals from N_r to N_w

Other possibilities (which are not determined by a pension system reform):

- Increase N_w : female labor participation, immigration...
- Increase w : higher productivity, economic growth

Incremental Reforms

Many possible additional reforms:

- Raise social security tax revenues collected.
 - Raise the tax rate on taxable wages.
 - Extend the base of taxable wages.
- Reduce social security benefits paid out.
 - Raise the retirement age to match increasing life spans.
 - Lower benefits to all eligible recipients.
 - Change to a welfare program where recipients' benefits decrease with higher incomes.

Fundamental Reforms

Fundamental reforms have also been proposed:

- Invest the trust fund in stocks
- Privatization

Advantages of a privately funded pension system with individual accounts:

- the capital is in the hands of individuals rather than the State;
- it respects consumer sovereignty over investment choices.
- returns are higher.

A crucial question: **how can the current generation of pensioners be supported?**

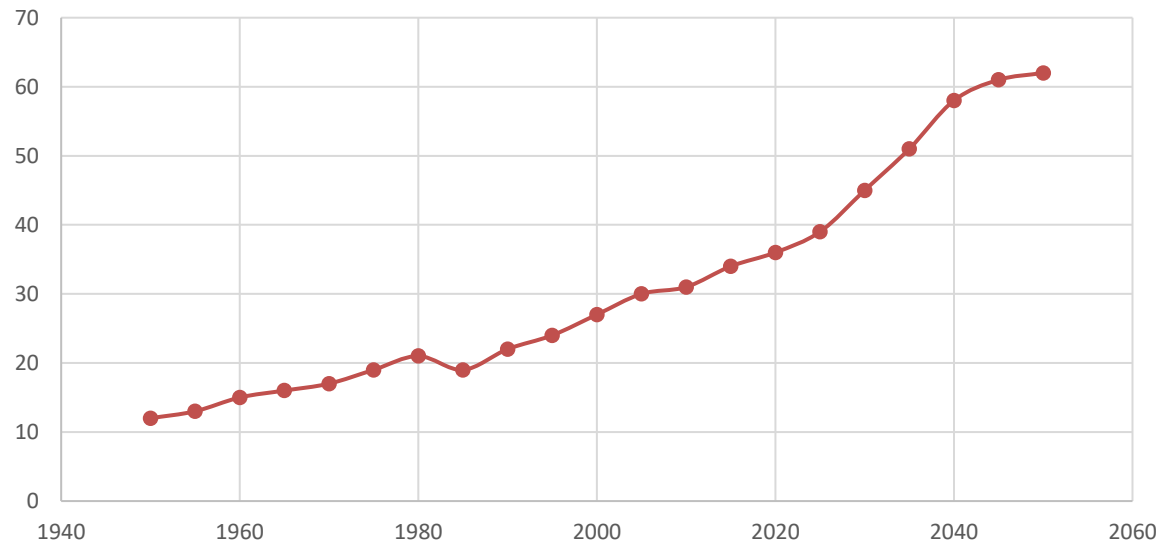
Other critical issues:

- Would a private pension system really offer a higher rate of return? The debt inherited from the first generation of pensioners would need to be financed.
- Administrative and management costs would be very high
- Policymakers might be unwilling to respect consumer sovereignty over savings earmarked for pensions.

A focus on the Italian case

The demographic dangers of pay-as-you-go pension systems

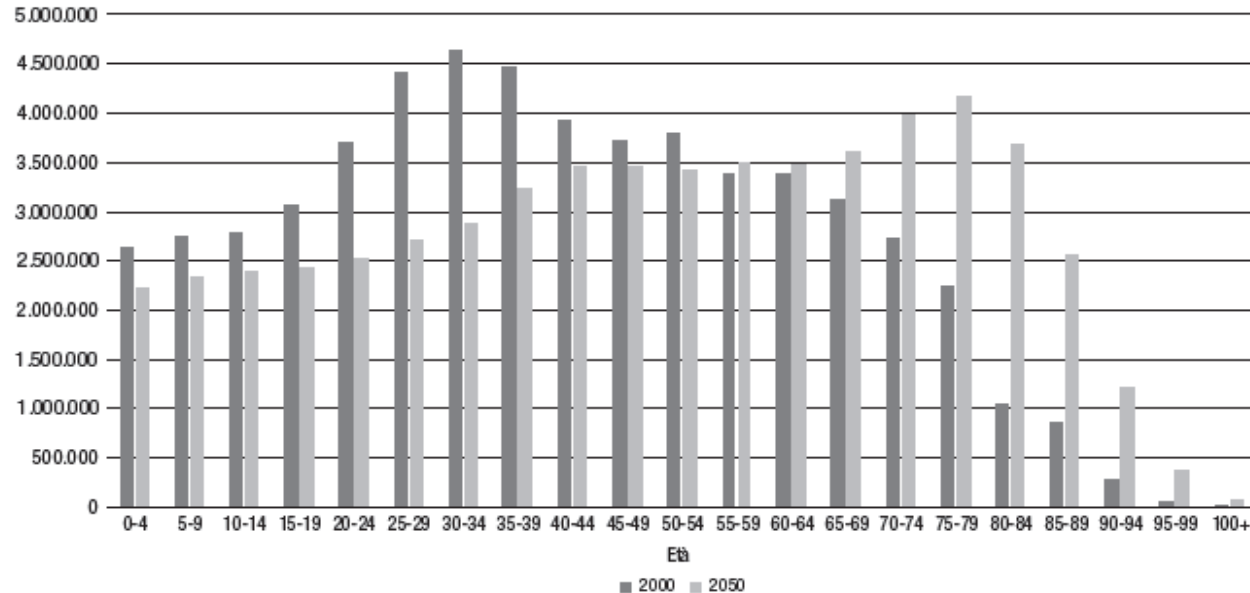
Italy - Elderly dependence index:
Population 65+ / Population 15-64



Population trends in Italy

Since the 1990s, the Italian pension system has been under strong pressure due to population aging, driven by rising life expectancy and declining birth rates.

Figura 9.8 La popolazione italiana per fasce di età (2000-2050)



A timeline of Italian reforms

Pre-Amato:

- earnings-related calculation method with pensionable earnings equal to the **average of earnings over the last five years**, $\beta = 2\%$.
- Pensions were indexed to both wage growth and inflation.

← 1992
Amato

← 1995
Dini

← 2011
Fornero

The Amato Reform (1992)

← 1992
Amato

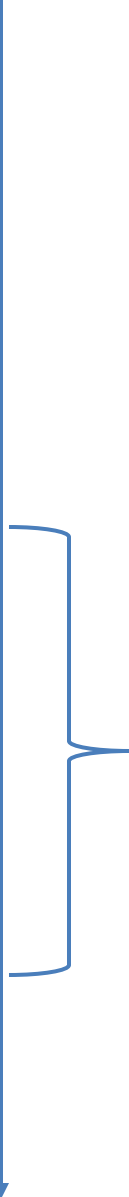
- The earnings-related calculation method is maintained, but pensionable earnings are no longer calculated as the average of the last five years of earnings; instead, they are calculated as **the average of earnings over the entire working life, appropriately revalued.**
- The statutory retirement age for old-age pensions (with at least 20 years of contributions) increases from 60 to 65 for men and from 55 to 60 for women.
- Eligibility for seniority pensions requires at least 35 years of contributions.
- Pensions are indexed only to inflation and no longer to wage growth: the pension received in the first year increases in subsequent years only in line with inflation.
- From an intergenerational equity perspective, those who were already retired are favored relative to younger cohorts.

The Dini Reform (1995)

1995
Dini

- Remaining within a pay-as-you-go system, a shift is **introduced from the earnings-related calculation method to the contribution-based calculation method**: the pension is calculated by multiplying the accumulated contribution amount (obtained by applying a 33 percent contribution rate to earnings) by a rate equal to the five-year moving average of the growth rate of nominal GDP. This amount is then multiplied by a conversion coefficient that ensures equality between the accumulated contributions and total pension benefits and that varies with life expectancy at retirement: the lower the life expectancy (and thus the older the retiree), the higher the pension. **This mechanism discourages individuals from retiring early. The contribution-based calculation method reduces the level of pensions.**
- The conversion coefficients are updated every ten years, in order to adjust pensions to changes in life expectancy and thus avoid financial imbalances due to population aging.
- The retirement age is set within a window between 57 and 65 years. However, the above-mentioned disincentive effect still applies, meaning that in terms of pension benefits **it is advantageous to postpone retirement.**
- Pensions continue to be indexed only to the inflation rate.
- This system applies only to individuals who started working from 1 January 1996 onwards. It does not apply to those who had more than 18 years of contributions as of that date; for them, a pro-rata or mixed system applies.

In between the Dini reform and the Fornero reform

- 
- Prodi (1998): the age requirements for seniority pensions are raised. The aim is to harmonize regulations across different categories of workers.
 - Maroni Law (2004): eliminates the retirement flexibility introduced by the Dini reform. From 1 January 2008, a single requirement is introduced: either 40 years of contributions, or 35 years of contributions and 65 years of age for men / 60 for women.
 - Prodi (2007): to make access to pensions more flexible, a quota system is introduced, linking eligibility for retirement to both chronological age and contributory seniority.
 - 2009: equalization of the retirement age for men and women.

The Fornero Reform (2011)

- Application of the **contribution-based calculation method for everyone** starting from 1 January 2012. Even for those who had more than 18 years of contributions in 1995, a pro-rata system applies, under which the pension is calculated using the earnings-related method up to 2011 and the contribution-based method from 2012 onwards.
- **Adjustments to life expectancy**, which determine eligibility requirements and conversion coefficients, are made **every two years** starting from 2019.
- **Increase in contribution rates** for both self-employed workers and employees.
- **Abolition of the quota system** for retirement age and introduction of **new criteria for access to old-age pensions and early retirement pensions**. For old-age pensions, at least 20 years of contributions are required, along with a statutory retirement age that gradually increases to reach 67 years for everyone by 2020. The pension benefit is set at no less than 1.5 times the amount of the social allowance. For early retirement pensions, a lower statutory age is allowed, but stricter contribution requirements apply, together with a penalty (on the earnings-related component) of 1 percent for each year of retirement before age 62, and 2 percent for each additional year beyond the first two years of early retirement. “Contribution-based” workers, instead, may access early retirement at age 63 with 20 years of effective contributions, provided that the monthly pension exceeds 2.8 times the social pension.
- Indexation of pensions to inflation is provided in 2012–13 **only for pensions below** three times the minimum benefit.

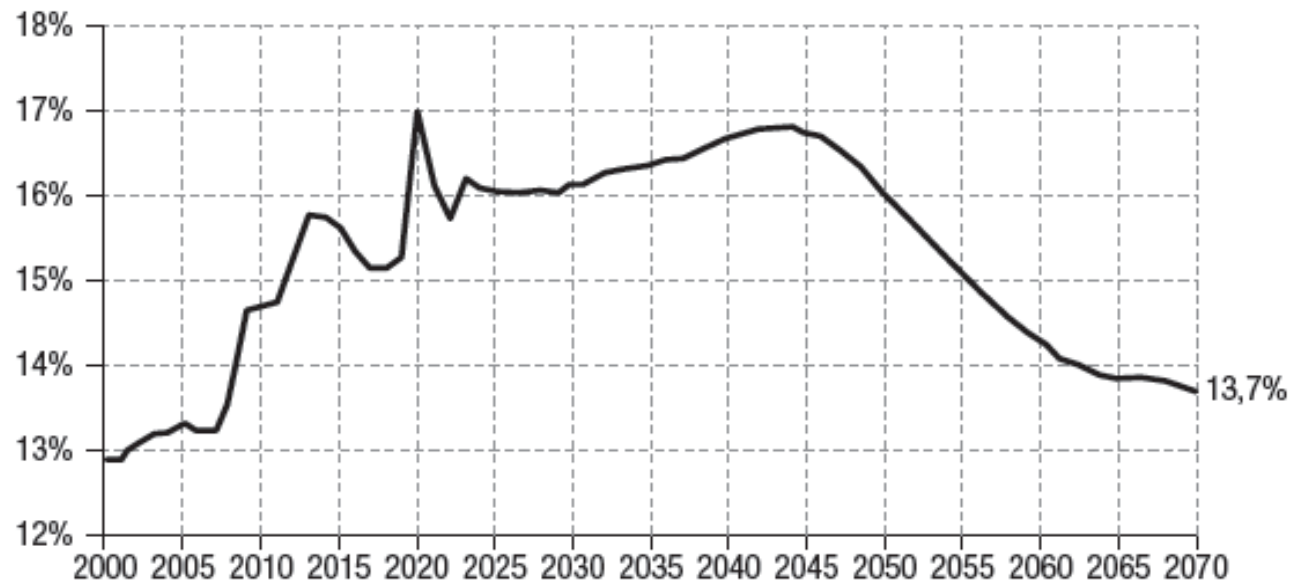


2011
Fornero

Flexibility in the retirement age

- The scenario following the Fornero Reform reopened the issue of flexibility in retirement: there are workers who wish to retire earlier and are potentially willing to accept a penalty.
- Examples: APE sociale, Quota 103, and Opzione Donna.

Public pension spending over GDP in Italy (2000-2070)



Exercises

1) In the state of Fantasyland, workers can choose between a pay-go system and a fully-funded system. Given that the expected interest rates are about 2%, the expected population growth rate is 3% and the expected wage growth rate is 1%, what do you expect to observe?

- a) All workers will opt for the pay-go system
- b) All workers will opt for the fully funded system
- c) 50% of workers will opt for the pay-go system and 50% of workers will opt for the fully funded system
- d) We cannot answer given the insufficient data provided

2) Consider a citizen who worked for three years before retiring, with salaries equal to 40, 45 and 50 respectively in the three years. Her pension gets computed using the earning-based method. If the coefficient of return is 0,2 percent, how much would her pension be?

- a) 30
- b) 50
- c) It depends on how the pensionable wage is computed (ex: lifetime average vs last year salary)
- d) It depends on the payroll tax rate alpha

3) Which of the following best characterizes the Dini Reform (1995) of the Italian pension system?

- a) It replaced the pay-as-you-go system with a fully funded system
- b) It introduced a contribution-based calculation method linked to life expectancy
- c) It indexed pensions to wage growth rather than inflation
- d) It immediately applied the new system to all workers regardless of seniority

4) Which of the following was a key effect of the Fornero Reform (2011)?

- a) It reintroduced retirement age flexibility through a quota system
- b) B. It abolished the contribution-based method for older workers
- c) It accelerated the increase in the statutory retirement age and linked it to life expectancy
- d) It restored pension indexation to both wages and inflation

5) Which statement best describes a key disadvantage of a pay-as-you-go (PAYG) pension system?

- a) It cannot provide pensions to the first generation of retirees
- b) Its return depends on financial market performance
- c) It is vulnerable to population aging and declining worker-to-retiree ratios
- d) It eliminates moral hazard in retirement decisions

True/False Questions (please provide an explanation)

- 1) In a pay-as-you-go pension system, the implicit rate of return is unaffected by demographic changes once the system is in financial equilibrium.
- 2) Transitioning from a pay-as-you-go system to a fully funded system can make all generations better off without additional financing costs.
- 3) In a contribution-based PAYGO pension system, linking benefits to life expectancy reduces the financial impact of population aging on the system.

Exercise 1

Consider a country with a Pay-as-you-go pension system, The average salary if workers is $w=200$;

- a) Assume that there are $N_r=2000$ retired people e $N_w= 3000$ workers and that the payroll tax rate α is 0,5, please compute the replacement rate (assuming no population growth and constant salaries)
- b) Assume 1000 workers, whose salary is equal to the average one, leave the country. What is the new payroll tax rate we would have in equilibrium, ensuring the same average pension as in point a?
- c) If we wanted to keep rate α equal to 0,5 even after the emigration, how much would the replacement rate then be?

Exercise 2

Consider an individual who worked and paid contributions for three period. Her wage was 20 in the first period, 15 in the second, 10 in the third.

- a) Please compute the pension to be received by this individual under the earning-based method, considering as pensionable wage the average wage over the full career, given a coefficient of return β of 0,05.
- b) How would your answer to the previous part change if the pension were to be computed using as pensionable wage the last wage received by the individual?